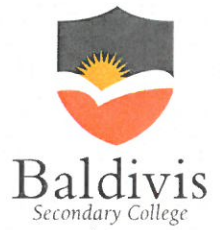


Year 10
Term Test 2 Pathway 1
Finance, Trigonometry, Statistics, Linear Quadratics



Resources allowed: Calculator and 1 x A4 page double side

Name: Solutions Class: _____ Time: 55 minutes Total Marks: ____/54

Show working and answers on this sheet. Show working in sufficient detail to support your answers.

Incorrect answers without supporting reasoning may be allocated zero marks.

1. [7 marks: 1, 1, 2, 3]

- a) Find the simple interest charged on \$50 000 borrowed for 3 years at 8.5% pa.

$$I = 50\,000 \times 0.085 \times 3$$
$$= \$12\,750 \checkmark$$

- b) How much would need to be invested at 6.5% pa for 6 months in order to earn \$500 interest

$$500 = P \times 0.065 \times 0.5$$
$$500/0.0325 = P$$
$$\$15\,384.62 = P \checkmark$$

- c) \$25 500 is invested at 5.3% pa compounded yearly. What will be the total amount after 4 years (round to the nearest dollar)?

$$A = 25\,500(1.053)^4 \checkmark$$
$$A = \$31\,351 \checkmark$$

- d) State which option produces the better outcome over 5 years? Show your working.

- i) \$100 000 invested at 9% pa compounded yearly?
ii) \$100 000 invested at 9% pa compounded monthly?

i) $A = 100\,000(1.09)^5$

$$A = \$153\,862.40 \checkmark$$

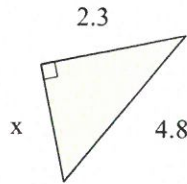
ii) $A = 100\,000\left(1 + \frac{0.09}{12}\right)^{5 \times 12}$

$$A = \$156\,568.10 \checkmark$$

Option ii $\checkmark$

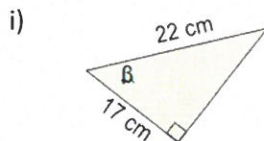
2. [14 marks: 1, 2, 2, 2, 1, 3, 3]

- a) Use Pythagoras' theorem to find the length of the unknown side (round to two decimal places):

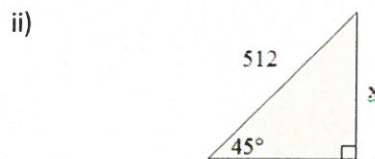


$$x = \sqrt{4.8^2 - 2.3^2} \\ = 4.21 \checkmark$$

- b) Use the sin, cos, or tan ratio to find the unknown (round to two decimal places):

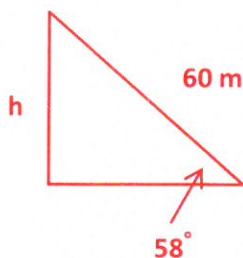


$$\beta = \cos^{-1} \frac{17}{22} \\ = 39.40^\circ \checkmark$$



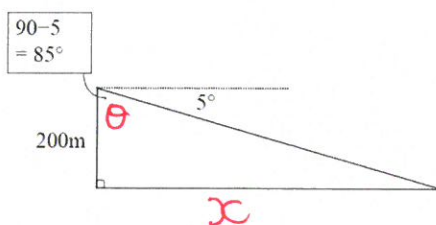
$$x = 512 \sin 45^\circ \\ = 362.04 \checkmark$$

- c) A television mast is secured by a single wire cable which is 60 metres long. The cable is fastened to the top of the mast and to the ground. It makes an angle of 58° with the ground. Find the height of the mast.



$$h = 60 \sin 58^\circ \checkmark \\ = 50.88 \text{ m} \checkmark$$

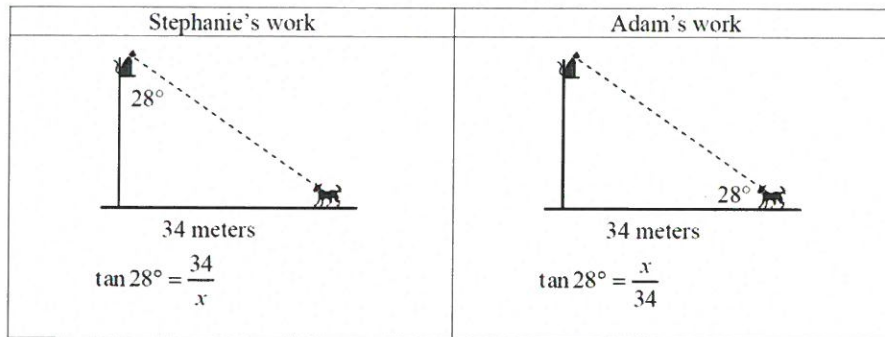
- d) From the top of a 200 m tower, the angle of depression to a fire is 5° . How far away is the fire?



$$x = 200 \tan 85^\circ \checkmark \\ = 2\,286.01 \text{ m} \checkmark$$

- e) A 4.5 m ladder rests against a tree on level ground and forms a 75° angle of elevation. Where is the correct location of the 75° angle?
- Between the ladder and the ground
 - Between the ladder and the tree
 - Between the tree and the ground
 - It is not possible to place a 75° angle on such a figure.

- f) Error Analysis: Consider the following problem, which Stephanie and Adam are both trying to solve: "A cat, who has climbed a tree, looks down at a dog at a 28° angle of depression. If the dog is 34 meters from the base of the tree, how high up is the cat?" The first steps of their work are shown below. Analyze their work and determine who, if anyone, has set it up correctly.



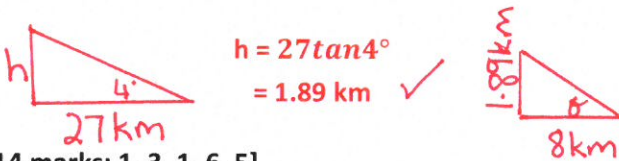
Stephanie's work is incorrect because she drew the 28° angle relative to the vertical tree. Adam's work is correct because he drew the 28° angle relative to the horizontal. Technically, Adam drew an angle of elevation of 28° , but because all horizontal lines are parallel, his placement of the angle and the equation he wrote will, in fact, yield the correct solution.



How high up in the tree is the cat?

The cat has climbed up approximately 18 meters. ✓

- g) A person starts out 27 km from the base of a tall mountain, and looks up at a 4° angle of elevation to the top of the mountain. When they move 19 km closer to the base of the mountain, what will be their angle of elevation when they look to the top? Answer to the nearest degree.



$$\theta = \sin^{-1} \frac{1.89}{8} = 13.29^\circ = 13^\circ \quad \checkmark \text{ The angle of elevation will be } 13^\circ. \quad \checkmark$$

3. [14 marks: 1, 3, 1, 6, 5]

- a) Which of these graphs would touch the x-axis but not cross it?

A $y = x^2 + 3x$

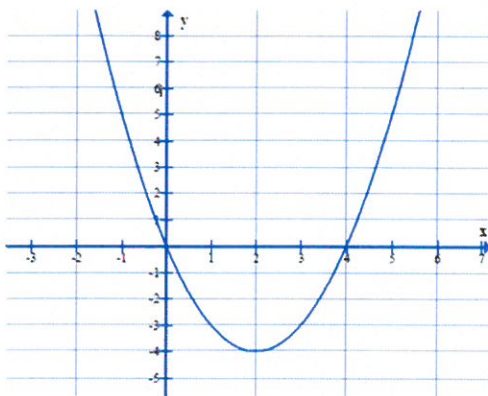
B $y = x^2 + 2x + 4$

C $y = x^2 + 10x + 20$

D $y = x^2 + 8x + 16$

- b) Answer the following:

i)



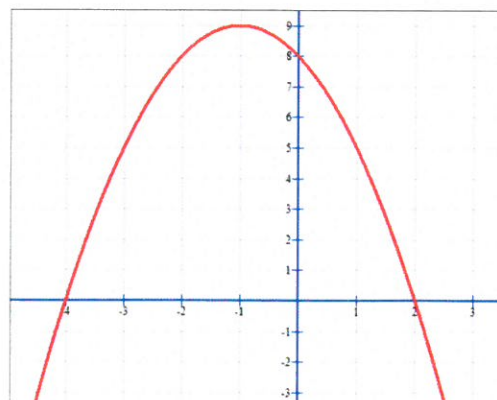
Equation of the axis of symmetry: $x = 2$

Coordinates of the turning point: $(2, -4)$

Maximum or Minimum? Minimum

✓1/2 each 3

ii)



Equation of the axis of symmetry: $x = -1$

Coordinates of the turning point: $(-1, 9)$

Maximum or Minimum? Maximum

✓1/2 each

c) The y-intercept of the graph of $y = -x^2 - 6$ is:

A 1 B -1 C 6 D -6 ✓

d) Complete the tables and then plot the points and join with a smooth curve on the grid provided.

$y = 2x^2$

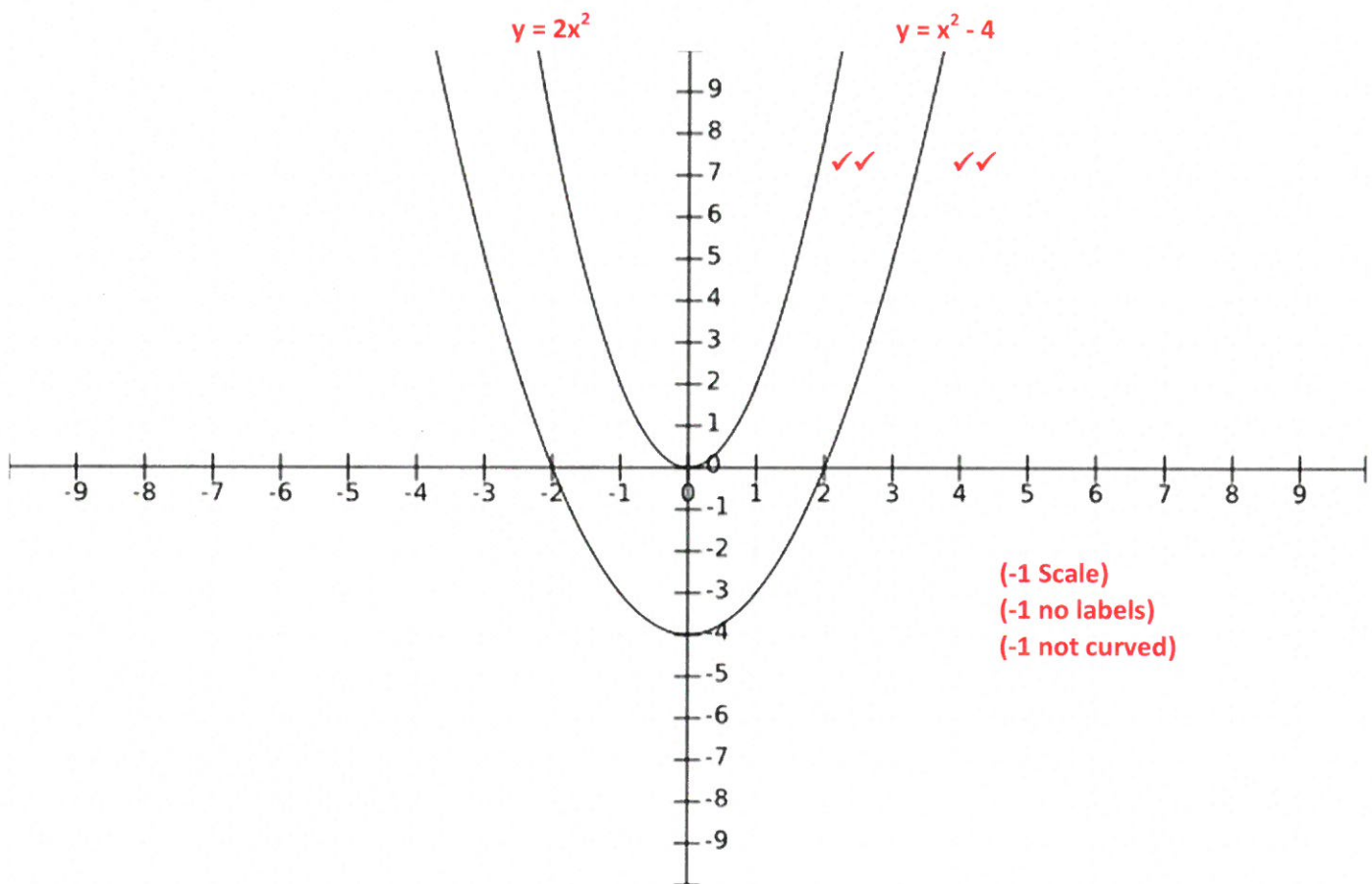
x	-3	-2	-1	0	1	2	3
$y = 2x^2$	18	8	2	0	2	8	18

$-\frac{1}{2}$ each error ✓

$y = x^2 - 4$

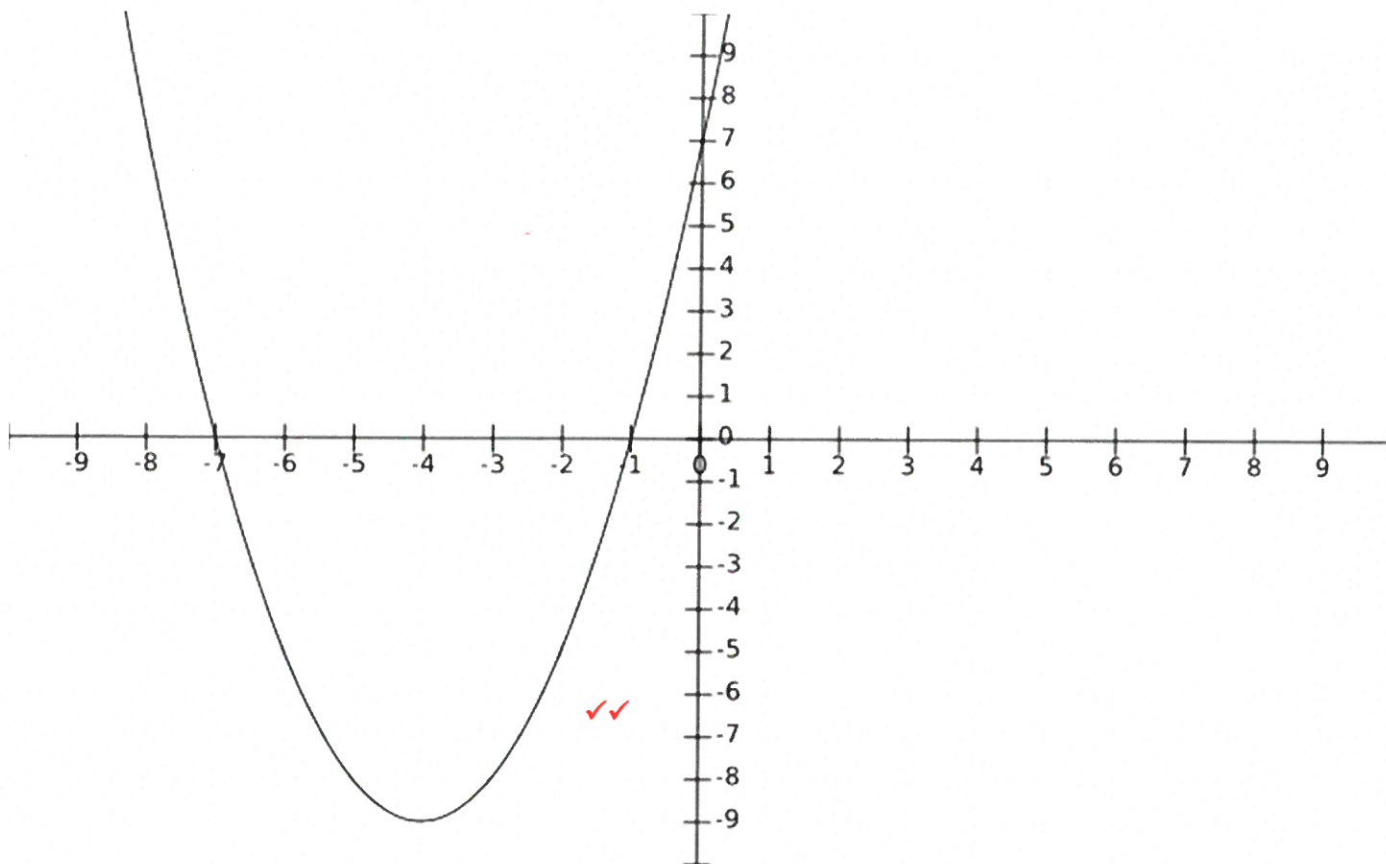
x	-3	-2	-1	0	1	2	3
$y = x^2 - 4$	5	0	-3	-4	-3	0	5

$-\frac{1}{2}$ each error ✓



- e) **Without** using a table of values, sketch the following parabola on the axis provided.
 $y = x^2 + 8x + 7$

(-1 Scale)



axis of symmetry: $x = -4$ ✓

Turning Point: $y = (-4)^2 + 8(-4) + 7$
 $= 16 - 32 + 7$
 $= -9$
 $(-4, -9)$ ✓

x- intercepts: $0 = (x + 1)(x + 7)$
 $x = -1, -7$
 $(-1, 0), (-7, 0)$ ✓

y-intercept: $(0, 7)$ ✓

4. [19 marks: 4, 3, 8, 4]

a) Draw the boxplot for the following set of data by first completing the 5 number summary.

Mathematics scores for a Year 10 end of Year Examination (%)

Scores	42	47	52	55	55	58	61	62	64	65	67	68	72	80	88	92
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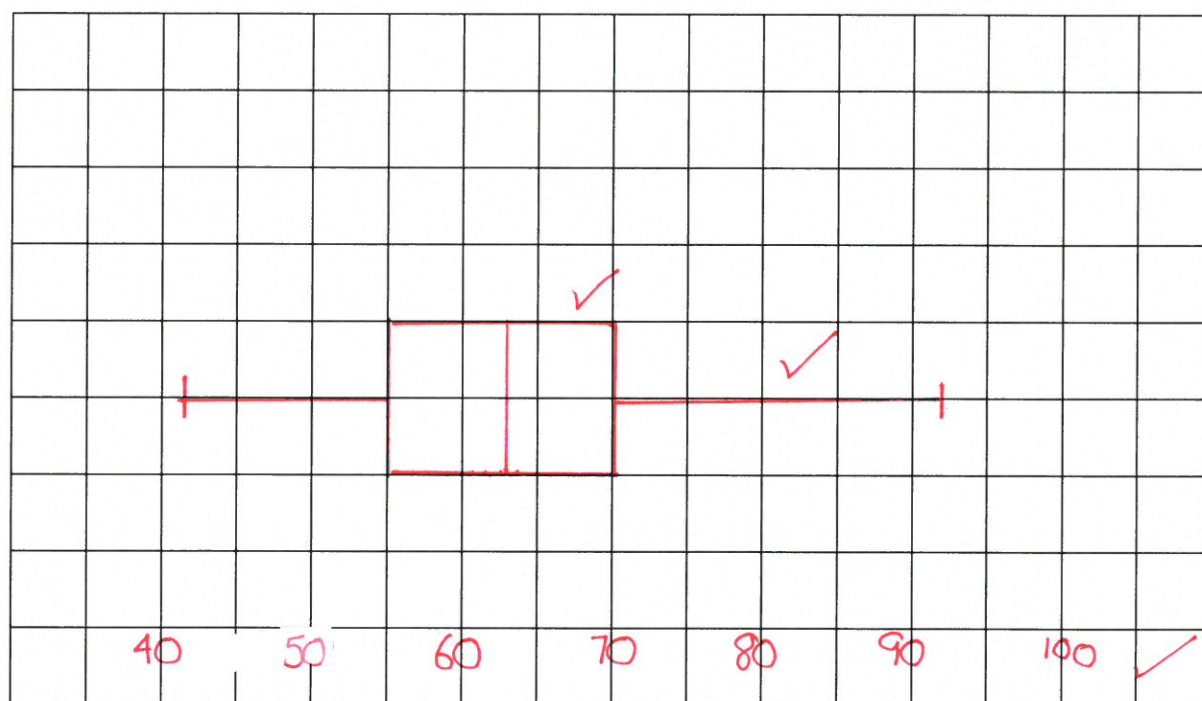
Lower = 42

Upper = 92

Q1 = 55

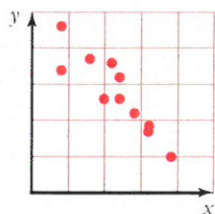
Q2 = 63

Q3 = 70



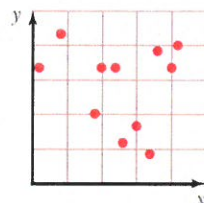
b) State whether any relationship is indicated by each of the following scatter plots, and if so, state the kind of relationship shown.

i)



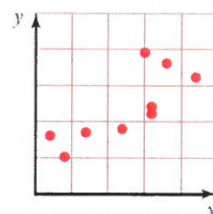
Strong negative ✓

ii)



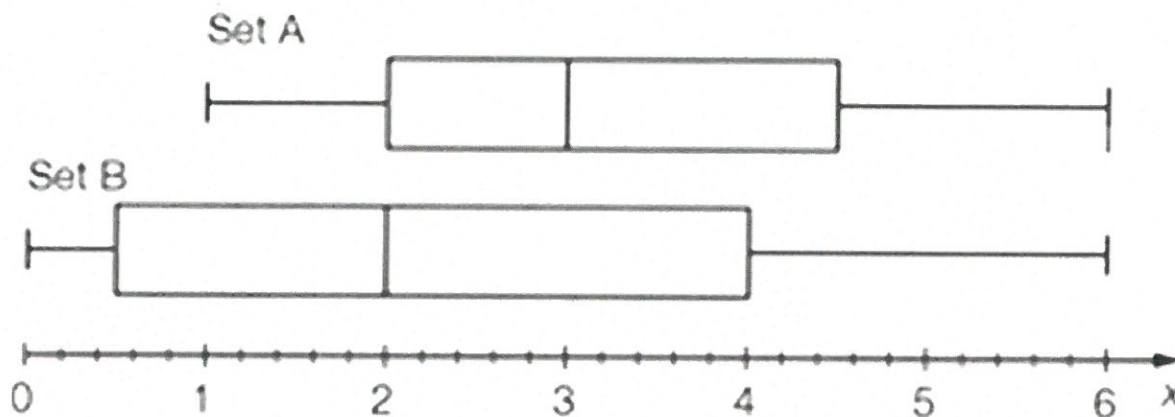
no correlation ✓

iii)



weak positive ✓

- c) The following boxplots are mathematics quiz scores, out of 8, for two classes, A and B.



- i) For each class find the five number summary and the interquartile range (IQR)

	Set A	Set B
Minimum	1	0
Q_1	2	0.5
Median	3	2
Q_3	4.5	4
Maximum	6	6
IQR	2.5	3.5

ii)

Compare the statistics for the two classes. Which did the best? Justify your answer.

Set A did best.

Minimum 2 valid points:

- Set A has a smaller range
- The middle 50% of Set A students achieved higher than median of set B
- Any other reasonable response.

- d) Calculate the mean and standard deviation of the following sets of data:

- i) 45, 58, 48, 54, 51, 57

mean = 52.17 ✓ SD = 4.67 ✓

ii)

Score	28	29	30	31	32	33	34	35	36	37	38	39
Frequency	2	3	1	3	1	4	6	3	6	14	3	4

mean = 34.9 ✓ SD = 3.04 ✓